Automotive Design 🚙

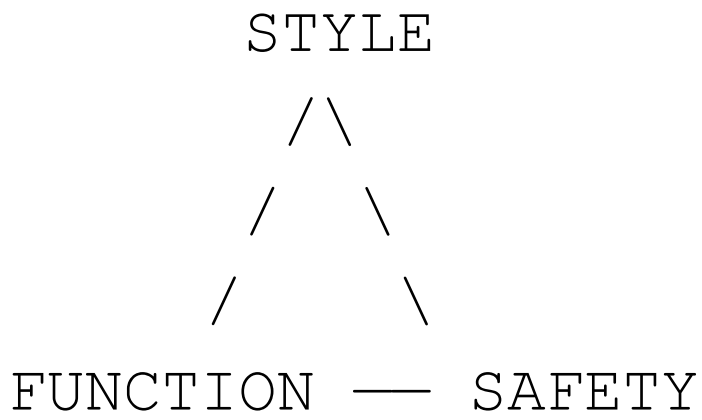
Junior Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

From idea to road 

The Automotive series follows the real industry pipeline:
Design it → **engineer** it → **manufacture** it
Today: design it for a **user** and a **purpose**.

Every design is a trade-off ⚖️



You **can’t max out all three**. Designers choose deliberately.

Which corner did each one pick? 🤔

Vehicle	Favors	Gives up
Sports car	style + speed	cargo space
Minivan	function	style
Delivery truck	function	style

Designers don’t pick everything — they choose for their user.

The Design Brief 📋

Your team draws a brief card. Unpack it like professionals:

- **Who** is the user?
- What must it **do**? *(function)*
- What would hurt them if it failed? *(safety)*
- What would make them **love** it? *(style)*

Write your requirements 

Three must-haves + one nice-to-have

That’s a real requirements list — with constraints.

Sketch TWO. Choose ONE.

Sketch **two different concepts** — side view + top view, features labeled.

Committing to your first idea is the rookie mistake.

Comparing two forces real design thinking.

Then pick one — and say what it trades away.

Build the concept model 🛠️

Cardboard and foam. Show **every must-have feature**:

- A ramp that folds
- A market stall that opens
- Weatherproof mail access

It doesn’t need to drive. It needs to **communicate the design**.

Design Review 🎤

Each team presents: 1. Your **user** 2. Your **must-haves** 3. The **trade-off** you chose 4. The feature you're **proudest** of

Audience: one "what about…?" question per team.

Defending your choices is the real job.

Robot moment 

Our team designs the robot under the same triangle every season — **speed vs. capability vs. not-breaking**.

Ask us what this year's robot traded away.

What you accomplished ✅

- Explained how style, function, and safety trade off
- Turned a user’s needs into a must-have requirements list
- Compared two concept sketches and justified your choice
- Built and presented a concept model of your design

Keep going at home! 🏠

- Pick any vehicle you see: which corner of the triangle did it choose?
- Sketch two concepts for anything — then choose, and say why
- Ask your troop leader about the next Automotive badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.